Enhancing Egyptian Arabic Multimodal Emotion Recognition

via Cross-Modal Attention

Can feature-level interaction between **text**, **audio**, and **video** improve Egyptian Arabic emotion recognition beyond gated late fusion?

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German International University - Research Project I - Winter 2026

Roadmap: From Problem to Experiment

1. Context

- How the literature was reviewed
- Multimodal emotion recognition
- Egyptian Arabic and sarcasm
- EGY-MER dataset constraints

2. Proposed method

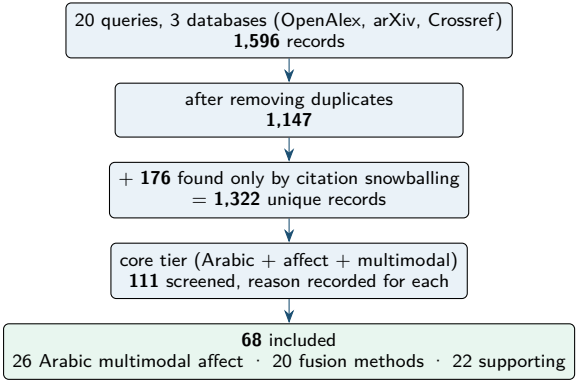
- Why late fusion is limited
- Tri-CAN architecture
- Cross-attention, LoRA, focal loss

3. Experiment

- Baselines and ablations
- Metrics and validation
- VAD-avatar idea as future work

The question is whether letting the modalities interact helps with Egyptian Arabic sarcasm and cases where words and delivery disagree.

How the Literature Was Reviewed



Protocol

- Fixed before searching; covers everything up to 31 July 2026
- Snowballing in both directions from 4 seed papers: EGY-MER, Ar-MuSA, UniTextFusion, MulT
- Every record and decision is in the tracking spreadsheet (`RP1_paper_tracking.xlsx`)

Limitations

- Screening at title and abstract level
- Semantic Scholar needed an API key we did not have

What the Review Found in 2026

Attention models on Arabic sentiment

Ar-MuSA, 3-class sentiment	F1
UniTextFusion (2025)	0.71
Tri-modal early fusion (2026)	0.776
GCMA-Net, gated cross-modal attention (2026)	0.802
Directional semantic enhancement (2026)	0.899

Arabic emotion: no follow-up yet

- All of this work is on Ar-MuSA and sentiment polarity.
- EGY-MER (7 emotions, Egyptian Arabic) has no published follow-up; its 0.519 baseline is untouched.
- New emotion corpora exist, e.g. MORAD: 2,325 Moroccan videos with the same 7 emotions.

This changed the fourth gap in the proposal: the mechanism is no longer new to Arabic, but the task is.

Khaled et al., 2025; Ben Cheikhi et al., 2026 (three papers); Atou et al., 2026

Motivation: Why Egyptian Arabic Multimodal Emotion?

Why multimodal?

Human emotion is not carried by text alone.

- **Text**: words and lexical sentiment
- **Audio**: pitch, tone, pauses, intensity
- **Video**: facial expression, eye movement, gestures

Recurring example

"Ah, momtaz!"
"Oh, excellent!"

Why Egyptian Arabic?

- Dialectal Arabic is under-resourced compared with English.
- Egyptian Arabic has informal spelling, code-switching, and cultural expressions.
- Sarcasm often creates **polarity contradiction**: positive words, negative intent.

Presentation intuition

The same phrase can be praise or criticism depending on flat tone, mocking voice, or an eye-roll.

Rahma et al., 2023; Hussein et al., 2025

Dataset Context: From Scarcity to EGY-MER

Arabic multimodal dataset landscape

AMMD	EGY-MER	Ar-MuSA
2019 40 videos / 830 utt. MSA + Gulf + Levantine	2025 363 segments Egyptian Arabic only	2025 8,700 samples Egyptian + MSA

Primary dataset: EGY-MER

- 363 clean single-speaker Egyptian Arabic segments
- Sources: talk shows and dramas
- Modalities: transcript, audio, video
- Labels: 7 emotion classes

Two practical constraints

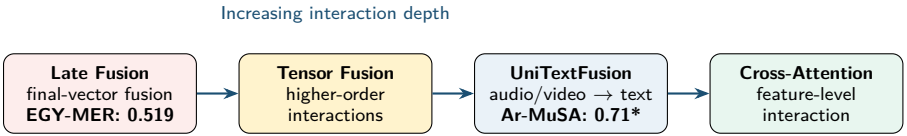
1. **Small data**: 363 samples is risky for large models.
2. **Class imbalance**: Fear and Surprise each constitute less than 4%.

Why this shapes the method

The proposal needs both a stronger fusion mechanism *and* a training strategy that avoids overfitting.

Alqarafi et al., 2019; Hussein et al., 2025; Khaled et al., 2025b

Fusion Techniques: Where the Gap Appears



What late fusion does well

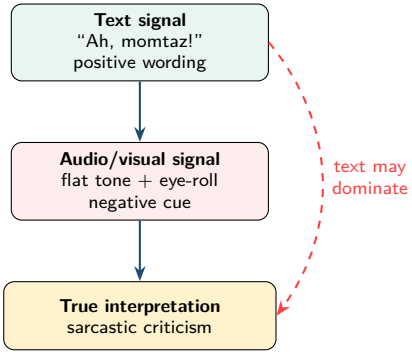
- Modular and easier to train
- Each encoder can be optimized independently

What it misses

A scalar fusion weight cannot ask: *which exact audio moment or facial frame explains this specific word?*

Hussein et al., 2025; Khaled et al., 2025a; Tsai et al., 2019. *Not directly comparable.

Why Late Fusion Struggles with Sarcasm



The independence assumption

Late fusion combines already-compressed modality vectors at the final decision stage.

Failure mode

If text is the strongest unimodal predictor, contradictory tone/face cues can be underused.

Research implication

The architecture should model **incongruence** between modalities, not only aggregate them.

Four Gaps Driving the Proposal

1. Independence assumption

Late fusion treats text/audio/video as mostly independent until classification.

2. Sarcasm bottleneck

Egyptian Arabic often relies on contradiction between positive wording and negative tone/face.

3. Data efficiency

EGY-MER has 363 samples across 7 classes, so full end-to-end training risks overfitting.

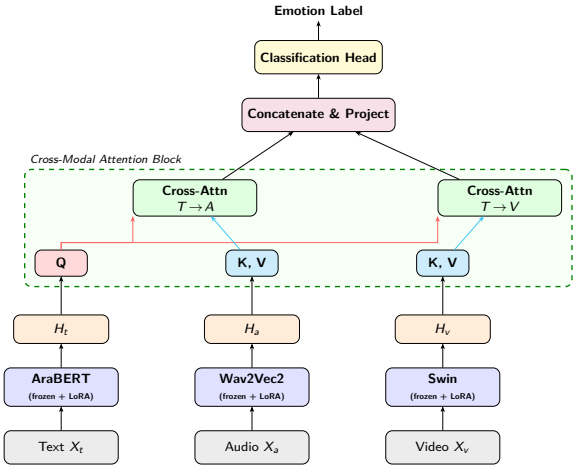
4. Arabic *emotion* recognition

Cross-modal attention reached Arabic sentiment in 2026, but only on Ar-MuSA's 3-class polarity. Nobody has tried it on 7-class Arabic emotion, where EGY-MER's baseline (0.519) is still unchallenged.

Tri-CAN is aimed at all four: modality interaction, sarcasm, small data, and emotion rather than sentiment.

Tsai et al., 2019; Hussein et al., 2025; Rahma et al., 2023; Ben Cheikhi et al., 2026

Proposed Architecture: Tri-CAN



Core idea

Text actively **queries** audio and video before classification.

Encoders

- Text: AraBERT
- Audio: Wav2Vec2
- Video: Swin Transformer

Difference from late fusion

Instead of fusing only final vectors, Tri-CAN performs feature-level alignment between words and non-verbal cues.

Cross-Attention Intuition: Queries, Keys, Values

Query (Q)	Keys (K)	Values (V)
The question being asked. **Here:** text asks which audio/video cues are relevant.	Searchable descriptors of audio/video moments. Examples: pause, shout, sarcastic laugh, eye-roll.	The actual information retrieved once relevance is known. These enrich the text representation.

Search-engine analogy

Query = what you type; Keys = searchable page titles/descriptors; Values = page contents returned.

Important nuance

Cross-attention is not merely "merging features"; it is one modality dynamically retrieving relevant information from another.

The Attention Equation

Scaled dot-product attention

$$\text{Attention}(Q, K, V) = \underbrace{\text{softmax}\left(\frac{\overbrace{QK^T}^{\text{similarity}}}{\sqrt{d_k}}\right)}_{\text{attention weights}} \underbrace{V}_{\text{retrieved information}}$$

For text-to-audio

$Q = H_t W_Q$ comes from text.
$K = H_a W_K$ and $V = H_a W_V$ come from audio.

Question: Which audio frames explain this word?

For text-to-video

Same equation, but audio features H_a are replaced by visual features H_v.

Question: Which visual frames explain this word?

Training Strategy: Frozen Encoders + LoRA

Why not fully fine-tune everything?

- AraBERT, Wav2Vec2, and Swin are large pretrained models.
- EGY-MER has only 363 samples.
- Full fine-tuning would be expensive and could memorize the dataset.

Frozen means

The original pretrained weights are kept fixed; the encoder acts as a strong feature extractor.

LoRA: small trainable correction

Instead of updating all weights W:

$$W' = W + \Delta W = W + BA$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll d$.

Why it fits this project

- Up to around 98% fewer trainable parameters
- More feasible on 12–16 GB VRAM
- LoRA ranks to test: $r \in \{4, 8, 16\}$

Khaled et al., 2025a

Handling Imbalance: Focal Loss

Problem

Standard cross-entropy can favor majority classes. In EGY-MER, rare classes such as Fear and Surprise may be ignored.

Focal loss

$$FL(p_t) = -\alpha_t(1 - p_t)^{\gamma} \log(p_t)$$

What the terms do

- p_t: model confidence for the correct class
- $(1 - p_t)^{\gamma}$: down-weights easy examples
- $\gamma = 2$: common focusing setting
- α_t: gives higher weight to rare classes

Plain English

The model spends less effort on easy Neutral examples and more effort on hard/minority emotion examples.

Experimental Setup: What Will Be Compared?

Datasets

- **Primary**: EGY-MER - 363 segments, 7 emotions
- **Optional warm-start**: Ar-MuSA - 8,700 Arabic multimodal sentiment samples
- Warm-start transfers modality representations, not the final classifier, because the label spaces differ.

Baselines

Model	Reported metric
AraBERT text-only	baseline
Wav2Vec2 audio-only	baseline
Swin video-only	baseline
Gated Late Fusion	W-F1 = 0.519
UniTextFusion	W-F1 = 0.71*
Tri-CAN	**TBD**

*Reported on Ar-MuSA, not directly comparable to EGY-MER.

The fair comparison is Tri-CAN against the EGY-MER gated late fusion baseline; the Ar-MuSA number is only for context.

Hussein et al., 2025; Khaled et al., 2025a,b

Ablations and Evaluation Metrics

Ablation studies

1. **Text+Audio vs Text+Video**: which non-verbal modality contributes more?
2. **Late Fusion vs Cross-Attention**: isolate the fusion mechanism.
3. **LoRA rank sensitivity**: $r \in \{4, 8, 16\}$.

Hypothesis

Audio may be especially useful for Egyptian Arabic because sarcasm is often expressed through tone and prosody.

Evaluation protocol

- Stratified 5-fold cross-validation
- Report mean $\pm$ standard deviation
- Preserve minority classes across folds

Metrics

- **Weighted F1**: overall performance
- **Macro F1**: minority-class performance
- **Confusion matrix**: which emotions get confused
- **Sarcasm sensitivity**: qualitative analysis of incongruent samples

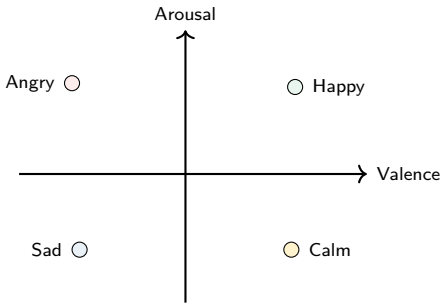
Alternative Direction Considered: Continuous Emotion Modeling

From classification to regression

Instead of predicting one of seven discrete labels, the model could predict a point in a continuous emotion space.

VAD emotion space

- **Valence**: positive vs negative
- **Arousal**: calm vs excited
- **Dominance**: in-control vs submissive



2D illustration; full VAD adds Dominance

Why it was interesting

Continuous emotion values could drive avatar facial expressions smoothly, instead of switching between fixed emotion presets.

Why the VAD-to-Avatar Idea Was Deferred

Potential value

- More expressive than seven fixed labels
- Supports emotion interpolation
- Useful for AR/VR avatars and rigged characters
- Connects affective computing to animation pipelines

Why out of scope for RP2

- Requires mapping discrete labels to VAD coordinates
- Changes learning objective from classification to regression
- Needs avatar rigging or animation prototype
- Evaluation likely needs user studies on naturalness/believability

We deferred this to keep RP2 to one measurable question: does cross-modal attention improve Egyptian Arabic emotion recognition?

Conclusion

Problem
Egyptian Arabic emotion is multimodal, sarcastic, and under-resourced.

Proposal
Tri-CAN replaces late fusion with text-driven cross-modal attention.

Experiment
Compare against baselines using 5-fold CV, ablations, and sarcasm analysis.

Thank you - Questions?

Nour Gaser & Mohamed Wael